Wind Speed / Gust Bug Fixes

WeeWX-Saratoga **rtcr.py** (RealtimeClientraw extension)

Background

This system uses WeeWX (weather station software) together with the **WeeWX-Saratoga** extension. A component of that extension called **rtcr.py** (“RealtimeClientraw”) reads live weather data and writes it, several times per minute, into a standard text file called **clientraw.txt**. Websites and dashboards built on the Saratoga/Weather Display template family read that file to display current conditions.

clientraw.txt has a well-known, publicly documented format: it's just a long line of numbers, and each position (“field”) in that line has a fixed, agreed-upon meaning — for example, field 4 is always outside temperature, field 6 is always barometric pressure, and so on. Every website and script built on this template family assumes those field meanings are correct.

Two separate bugs were found in **rtcr.py** that caused wind speed and gust values to display incorrectly. Both have been fixed.

Bug #1: Gust fields always matched current wind speed

Symptom

On the website dashboard, “gust” always showed the exact same number as the current wind speed, no matter how the wind was actually behaving. A real gust would never show as higher than the current reading and hold there — the two numbers moved in perfect lockstep, every time.

Root cause

rtcr.py calculates gust values (the highest wind speed seen in the last 1 minute, 5 minutes, or hour) using a function called `history_max()`. This function is supposed to look back through a list of recent wind readings and find the **highest speed** recorded. Instead, due to a one-character coding error, it was finding the reading with the **most recent timestamp** — which, by definition, is always just the current reading. So every “gust” calculation was silently returning “whatever the wind speed is right now” instead of a genuine maximum.

The fix

```
# Before (bug):
_max = max(snapshot, key=itemgetter(1))

# After (fixed):
_max = max(snapshot, key=itemgetter(0))
```

Fields affected: field 71 (max gust today), field 133 (max gust in the last hour), and field 140 (max gust in the last minute). All three now correctly report a real rolling maximum instead of mirroring current wind speed.

Bug #2: Field 2 was computed as a gust value, not current windspeed

Symptom

Even after fixing Bug #1, the dashboard's “Wind” reading (pulled from clientraw.txt field 2) still updated only slowly, holding the same value for long stretches, rather than reflecting the live, moment-to-moment wind speed a user would expect to see.

Root cause

According to the official, published clientraw.txt field specification, field 2 is documented as **“Current Windspeed.”** However, rtcr.py's own code computed field 2 as a **5-minute rolling maximum** (i.e., a gust value), not as the current reading — a mismatch between this code and the documented file format it's supposed to follow. This went unnoticed for a long time because Bug #1, above, coincidentally made this field behave *as if* it were showing current speed; once Bug #1 was fixed, this second, separate problem became visible.

The fix

```
# Before (bug — computed a 5-minute max instead of current speed):
if self.gust_period > 0:
    _gust = self.buffer['windSpeed'].history_max(packet_wx['dateTime'],
                                                  age=self.gust_period).value
else:
    _gust = self.buffer['windSpeed'].last

# After (fixed — always the current reading, per spec):
_gust = self.buffer['windSpeed'].last
```

Net result

Field	Meaning (per official spec)	Status
1	60-second average windspeed	Unaffected, working as before
2	Current windspeed	Fixed — now a true live reading
71	Max wind gust today	Fixed — now a true daily maximum
133	Max gust in the last hour	Fixed — now a true hourly maximum
140	Max gust in the last minute	Fixed — now a true 1-minute maximum

Wind speed and gust are now two genuinely independent, correctly calculated values, matching both the intended behavior of the software and the official documented file format that all dashboards and scripts built on this template family rely on.

Who this affects

This is a fix to the shared **rtcr.py** extension code itself, not a site-specific patch — anyone else running an unmodified copy of the WeeWX-Saratoga RealtimeClientraw extension is affected by both bugs in the same way, and would see the same improvement if they applied the same two code changes.